

Gabriel Akerman

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EDUCATION

University of Waterloo

Bachelor of Applied Science in Electrical Engineering

Waterloo, ON

Expected Apr. 2031

Stephen Lewis Secondary School

Ontario Secondary School Diploma

Thornhill, ON

Jun. 2026

- Class of 2026 Valedictorian, YRDSB Student Trustee Award for Character

TECHNICAL SKILLS

Languages: Python, C++, C#

Hardware & Prototyping: Arduino, Raspberry Pi, Fusion 360, Soldering, Altium Designer, Oscilloscopes, LTSpice

Developer Tools & Software: Git, OpenCV, MediaPipe, Unity, MonoGame

PROJECTS

UW Orbital – Satellite Current-Sensing PCB | *Altium Designer, Circuit Design, PCB Layout, Soldering*

- Designed a high-side/low-side current-sensing PCB to detect abnormal satellite bus current (0–200mA).
- Implemented differential RC low-pass filters and decoupling capacitors to attenuate common-mode noise.
- Configured a TI INA180 amplifier, matching a 0.2-ohm shunt resistor to optimize microcontroller ADC range.
- Assembled and hand-soldered the fabricated board, bringing the design from schematic to functional hardware.

Eye-Tracking Laser Turret | *C++, Python, Arduino, OpenCV, MediaPipe, Microcontrollers*

- Built a 2-axis laser turret with dual servos that locates and tracks a user's eyes via webcam in real time.
- Applied MediaPipe's face mesh model to extract eye landmark coordinates from each video frame
- Converted eye coordinates to servo angles in Python, sent to Arduino via serial in real time.
- Tuned Arduino PWM signal timing and servo response to reduce jitter and minimize tracking latency.

Morse Code Audio & Visual Transmitter | *C++, Arduino, Embedded Systems*

- Built a device that classifies button-press timing into dots and dashes, decoding real-time Morse input into text.
- Classified button-press duration into dots and dashes, with debouncing to prevent false triggers.

Out of [CTRL] | *C#, MonoGame, Game Design*

- Designed and published a 2D puzzle game where players rewrite rules mid-level to solve puzzles.
- Built a rule-manipulation engine with dynamic state-checking algorithms to validate rule changes in real time.
- Implemented a tile-based collision system supporting real-time rule changes without breaking game state.

EXPERIENCE

Sales Associate

Jul. 2026 – Sep. 2026

H&M

Vaughan, ON

- Delivered high-volume customer service and supported daily floor operations in a fast-paced retail store.
- Managed product restocking, inventory handling, and point-of-sale transactions.
- Collaborated with floor team members to optimize visual merchandising and customer flow.

Track Officials Associate

Oct. 2023 – Sep. 2025

Athletics Ontario

Toronto, ON

- Officiated regional track meets, enforcing competition rules and safety guidelines.
- Managed event equipment, timing logistics, and crowd movement under strict time schedules.

LEADERSHIP & AWARDS

DECA Chapter President

Sep. 2023 – Jun. 2026

Stephen Lewis Secondary School

Thornhill, ON

- Progressed from Member to Vice President and Chapter President, leading training for 200+ students.
- Secured 1st place in regional exams and roleplay scenarios, qualifying for provincial competitions.

Varsity Swim Team Founder & Captain

Sep. 2024 – Jun. 2026

Stephen Lewis Secondary School

Thornhill, ON

- Founded and captained the high school swim team, managing practice schedules and team logistics.

Design & Research Competitions

Jul. 2024 – May 2026

SHAD Canada & University of Toronto Student Research Initiative (UTSRI)

Ontario, CA

- Won 1st Place at SHAD Canada for pitching an eco-friendly alternative road salt solution.
- Earned 1st Place Senior Science for road salts and 3rd Place in Chemistry for glowstick alternatives at UTSRI.